

JAVA INHERITANCE – QUESTION 3

Implementation of Inheritance Using an Interface

1. Objective

To implement inheritance using an interface in Java, demonstrate how a class implements an interface, and verify the program using multiple test cases.

2. Task Description

Implement all types of inheritance and provide appropriate explanations as code comments. For this question, the program demonstrates inheritance through interface implementation. The complete code should be maintained in the respective GitHub repository along with this PDF, and the program should be executed and verified using multiple test cases.

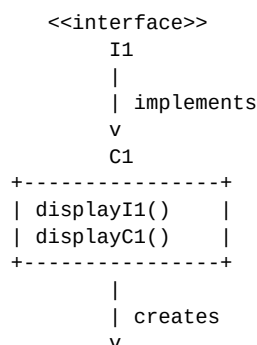
3. Explanation

- **I1** is an interface containing the abstract method **displayI1()**.
- **C1 implements I1** means class C1 follows and implements the contract defined by interface I1.
- Class C1 provides the implementation of **displayI1()**.
- **displayC1()** is an additional method belonging to class C1.
- The object **obj** is created using C1, and both methods are called.

4. Algorithm

1. Start the program.
2. Declare an interface I1.
3. Declare the abstract method displayI1() inside I1.
4. Create class C1 and implement interface I1.
5. Define displayI1() inside C1.
6. Define displayC1() inside C1.
7. Create an object of C1 in the main() method.
8. Call displayI1().
9. Call displayC1().
10. Stop the program.

5. UML Diagram



Question3
main()

6. Program Code

```
// Interface I1 represents the parent interface
interface I1 {

    // Abstract method declared inside the interface
    void displayI1();
}

// Class C1 implements interface I1
// This demonstrates inheritance using an interface
class C1 implements I1 {

    // Providing implementation for the interface method
    public void displayI1() {
        System.out.println("This method is implemented from I1");
    }

    // Method belonging to class C1
    void displayC1() {
        System.out.println("This is Class C1");
    }
}

// Main class
public class Question3 {

    public static void main(String[] args) {

        // Creating an object of class C1
        C1 obj = new C1();

        // Calling the method inherited from interface I1
        obj.displayI1();

        // Calling the method of class C1
        obj.displayC1();
    }
}
```

7. Expected Output

This method is implemented from I1
This is Class C1

8. Multiple Test Cases and Verification

Test Case	Operation	Expected Result
1	obj.displayI1()	This method is implemented from I1
2	obj.displayC1()	This is Class C1
3	Execute both methods	Both messages are displayed correctly

9. Result

The Java program was successfully executed. The interface I1 was implemented by class C1, and the interface method and class method were successfully invoked. The expected output was verified.

10. Submission Checklist

- Java source file: Question3.java
- PDF containing code, explanation, UML diagram, output, and test-case verification
- Output screenshot to be captured after execution and added to the final submission

- Both files to be pushed to the respective GitHub repository